



Shore School

2016
Year 12 Trial HSC
Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11–14, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Exam Number:

Set:

Total marks – 70

Section I Pages 2–5

10 marks

- Attempt questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–13

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

1 Which of the following does NOT have an inverse function?

(A) $y = \sqrt{x}$

(B) $y = x^2$

(C) $x = \sqrt{y}$

(D) $x = y^2$

2 What is $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{5x}$?

(A) $\frac{1}{15}$

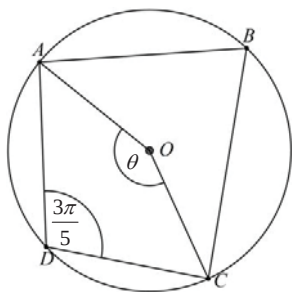
(B) $\frac{3}{5}$

(C) $\frac{5}{3}$

(D) 15

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

- 3 $ABCD$ is a cyclic quadrilateral, with all vertices lying on a circle with centre O .
The size of $\angle ADC$ is $\frac{3\pi}{5}$ radians.



NOT TO
SCALE

What is the size of angle θ ?

- (A) $\frac{\pi}{5}$
(B) $\frac{2\pi}{5}$
(C) $\frac{3\pi}{5}$
(D) $\frac{4\pi}{5}$
- 4 The point P divides the interval from $A(4, -2)$ to $B(6, 2)$ externally in the ratio $5 : 3$.
What are the coordinates of P ?
- (A) $(1, -8)$
(B) $\left(\frac{19}{4}, -\frac{1}{2}\right)$
(C) $\left(\frac{21}{4}, \frac{1}{2}\right)$
(D) $(9, 8)$

- 5 The parametric form of a parabola is $(3t, -6t^2)$.
What is the Cartesian form of this parabola?

- (A) $x^2 = -12y$
(B) $x^2 = -\frac{3}{2}y$
(C) $x^2 = \frac{3}{2}y$
(D) $x^2 = 12y$

- 6 A family of 8 sit at a circular table.
What is the probability that the two youngest children **do not** sit next to each other?

- (A) $\frac{5}{7}$
(B) $\frac{3}{4}$
(C) $\frac{20}{21}$
(D) $\frac{2519}{2520}$

- 7 Suppose θ is the acute angle between the lines $y = 3x - 1$ and $x + 2y - 6 = 0$.
What is the value of $\tan \theta$?

- (A) -7
(B) -1
(C) 1
(D) 7

8 What is the domain of $y = \cos^{-1}\left(\frac{3x}{2}\right)$?

(A) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

(B) $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$

(C) $-\frac{2}{3} \leq x \leq \frac{2}{3}$

(D) $-\frac{3}{2} \leq x \leq \frac{3}{2}$

9 When the polynomial $P(x) = ax^3 - 5x^2 + 10x + 12$ is divided by $(x - 2)$ the remainder is 4.

What is the remainder when $P(x) = ax^3 - 5x^2 + 10x + 12$ is divided by $(x + 3)$?

(A) -36

(B) -30

(C) -4

(D) -1

10 If $y = \cos(\ln x)$, which of the following is the correct expression for $\frac{d^2y}{dx^2}$?

(A) $\frac{d^2y}{dx^2} = -\cos(\ln x)$

(B) $\frac{d^2y}{dx^2} = \frac{-\cos(\ln x)}{x^2}$

(C) $\frac{d^2y}{dx^2} = \frac{\sin(\ln x) - \cos(\ln x)}{x^2}$

(D) $\frac{d^2y}{dx^2} = \frac{-\sin(\ln x)}{x}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Consider the polynomial $P(x) = 6x^3 - 5x^2 - 13x + 12$, with roots α , β and γ . 2

Determine the value of $\alpha^2 + \beta^2 + \gamma^2$.

(b) How many arrangements of the letters of the word PARRAMATTA are possible? 2

(c) Solve the inequality $\frac{5}{x-4} > x$. 3

(d) Use the substitution $u = e^x$ to evaluate $\int_0^{\ln \frac{1}{\sqrt{2}}} \frac{5e^x}{\sqrt{1-e^{2x}}} dx$. 3

Leave your answer in exact form.

(e) Differentiate $x^2 \cos^{-1} x$. 2

(f) Consider the binomial expansion of $\left(2x^3 - \frac{3}{x}\right)^{12}$. 3

Find an expression for the term independent of x .

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Sean took a piece of beef out of his freezer which was set at -20°C and placed it straight into an oven which was preheated to 160°C . The temperature at the centre of the meat may be modelled using the formula

$$T = 160 - Ae^{-kt},$$

where A and k are constants, and t is the time in minutes after the piece of beef was put in the oven.

- (i) After 50 minutes, Sean used his thermometer to discover that the beef was 40°C at its centre.

2

Show that $k = \frac{1}{50} \ln\left(\frac{3}{2}\right)$.

- (ii) Sean wants his beef to be cooked 'medium rare' which requires it to be 60°C at its centre.
How long after he put the beef in the oven, should Sean take it out?
Give your answer correct to the nearest minute.

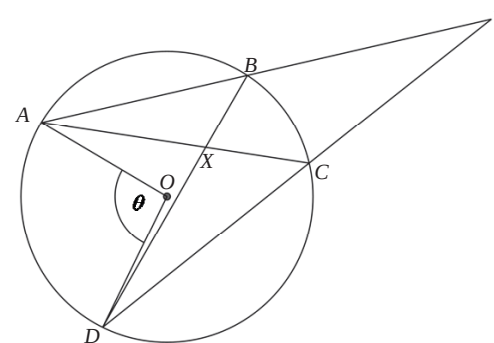
2

- (b) Ted tosses **two** unbiased coins in 10 separate trials.
Write an expression for the probability of both coins landing 'heads' in exactly 6 of these trials.

2

Question 12 continues on page 8

- (c) In the diagram, A, B, C and D are points on a circle centre O , and $\angle AOD = \theta$.
The lines AB and DC intersect at Y and the lines AC and DB intersect at X .



- (i) Explain why $\angle DBY = 180^{\circ} - \frac{\theta}{2}$.

1

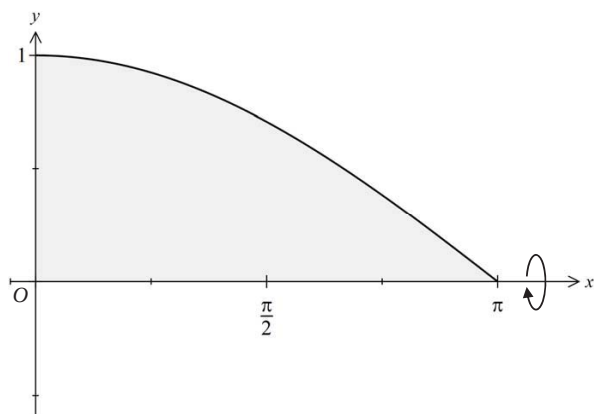
- (ii) Show that $\angle BXC + \angle BYC = \theta$.

2

Question 12 continues on page 9

- (d) The region bounded by the curve $y = \cos \frac{x}{2}$, the x -axis, and the y -axis is rotated about the x -axis to form a solid.

3



Find the volume of this solid of revolution.

- (e) Prove by mathematical induction that for all integers $n \geq 2$,

3

$$n^2 \geq 2n.$$

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the even function $f(x) = \frac{2x^2 + 9}{x^2 + 4}$.

- (i) Prove that $f(x) = 2 + \frac{1}{x^2 + 4}$. 1
- (ii) Hence, or otherwise, state the equation of the horizontal asymptote of the curve $y = f(x)$. 1
- (iii) Find the coordinates of the stationary point on $y = f(x)$. 1
- (iv) Hence, sketch $y = f(x)$, showing the features found above. 2
- (v) Find the area bounded by the curve $y = f(x)$, the line $x = -2$, the line $x = 2$, and the x -axis. 2

- (b) A particle is moving in a straight line according to the equation

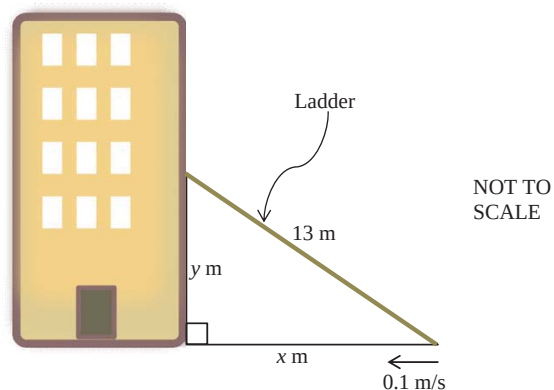
$$x = \sqrt{3} \sin 2t + \cos 2t$$

where x is the displacement in metres and t is the time in seconds.

- (i) Determine the value of R and α , for which $x = R \sin(2t + \alpha)$, 2
with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
- (ii) Prove that the particle is moving in simple harmonic motion by showing that x satisfies an equation of the form $\ddot{x} = -n^2 x$. 3
- (iii) Find when the particle is first stationary and its displacement at that time. 2
- (iv) Hence, or otherwise, find the distance travelled by the particle in the first π seconds. 1

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^3 = \cos x$ has only one solution, which is near $x = 0.9$.
Use one application of Newton's method to find a better approximation to the solution.
Leave your answer correct to 2 decimal places. 2
- (b) Find the general solution(s) of $4 \sin \frac{x}{2} \cos \frac{x}{2} = \sqrt{3}$.
Leave your answer in radians. 2
- (c) A fireman placed his 13 metre long ladder against a wall as shown in the diagram below. 2



As it didn't reach up high enough, he started pushing the foot of the ladder closer to the wall at the rate of 0.1 m/s.

Find the rate at which the ladder is moving higher on the wall when the foot of the ladder is 5 metres out from the base of the wall.

Question 14 continues on page 12

- (d) The equations of motion of a projectile fired from the origin with initial velocity $V \text{ ms}^{-1}$ at angle θ to the horizontal are

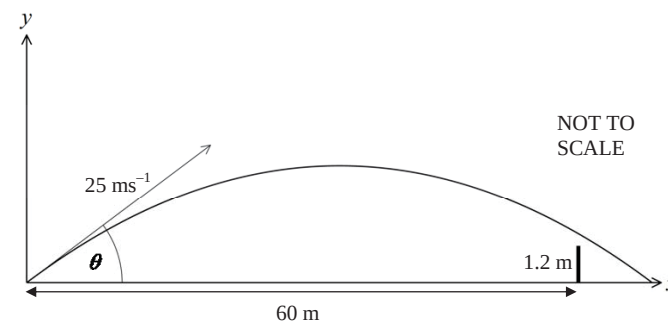
$$x = Vt \cos \theta \quad \text{and} \quad y = -\frac{1}{2}gt^2 + Vt \sin \theta. \quad (\text{Do NOT prove this.})$$

A cricket ball was hit from a point on horizontal ground at a speed of 25 ms^{-1} .

You may assume that $g = 9.8 \text{ ms}^{-2}$.

- (i) Prove that the flight path of the cricket ball is given by 2

$$y = -\frac{49x^2}{6250}(1 + \tan^2 \theta) + x \tan \theta.$$
- (ii) The cricket ball cleared a 1.2 metres high boundary fence 60 metres from where the ball was hit. 3



Determine the possible values of θ .

Question 14 continues on page 13

(e) Consider the binomial expansion

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + {}^nC_4x^4 + \dots + {}^nC_{n-1}x^{n-1} + {}^nC_nx^n.$$

(i) Find an expression for **1**

$${}^nC_0x^{-1} + {}^nC_1 + {}^nC_2x + {}^nC_3x^2 + {}^nC_4x^3 + \dots + {}^nC_{n-1}x^{n-2} + {}^nC_nx^{n-1}.$$

(ii) By differentiating, prove that **3**

$${}^nC_2 + 2{}^nC_3 + 3{}^nC_4 + \dots + (n-2){}^nC_{n-1} = (n-2)(2^{n-1} - 1).$$

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Yr 12 Maths Ext 1 Trial Solutions 2016

① B

$$y = \sqrt{x} \quad \nwarrow \nearrow \quad y = x^2 \quad \nwarrow \nearrow$$

$$x = \sqrt{y} \quad \nwarrow \nearrow \quad x = y^2 \quad \nwarrow \nearrow$$

② A

$$\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{5x} = \frac{1}{15} \times \lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{\frac{x}{3}}$$

$$= \frac{1}{15} \times 1$$

$$= \frac{1}{15}$$

③ D

$$\angle ABC = \pi - \frac{3\pi}{5} \quad (\text{opposite } \angle\text{'s of cyclic quadrilateral are supplementary})$$

$$= \frac{2\pi}{5}$$

$$\theta = 2 \times \frac{2\pi}{5} \quad (\angle \text{ at centre twice } \angle \text{ at circumference standing on same arc})$$

$$= \frac{4\pi}{5}$$

④ D

$$A(4, -2) \quad B(6, 2)$$

$$S: -3$$

$$P = \left(\frac{5(6) - 3(4)}{5 - 3}, \frac{5(2) - 3(-2)}{5 - 3} \right)$$

$$= (9, 8)$$

⑤ B

$$\begin{cases} x = 3t & (1) \\ y = -6t^2 & (2) \end{cases}$$

$$t = \frac{x}{3} \quad (1)$$

sub (1) into (2)

$$y = -6\left(\frac{x}{3}\right)^2$$

$$y = -\frac{2}{3}x^2$$

$$x^2 = -\frac{3}{2}y$$

⑥ A

$$P(\text{youngest not together}) = 1 - P(\text{youngest together})$$

$$= 1 - \frac{6! \times 2!}{7!}$$

$$= \frac{5}{7} \quad \text{Note } (n-1)! \text{ for circular arrangements}$$

⑦ D

$$y = 3x - 1 \quad m_1 = 3$$

$$x + 2y - 6 = 0 \rightarrow y = -\frac{1}{2}x + 3 \quad m_2 = -\frac{1}{2}$$

$$\tan \theta = \left| \frac{3 - (-\frac{1}{2})}{1 + 3 \times (-\frac{1}{2})} \right|$$

$$= \left| \frac{3.5}{-0.5} \right|$$

$$= 7$$

⑧ C

$$y = \cos^{-1}x \quad -1 \leq x \leq 1$$

$$\therefore y = \cos^{-1}\left(\frac{3x}{2}\right) \quad -1 \leq \frac{3x}{2} \leq 1$$

$$-\frac{2}{3} \leq x \leq \frac{2}{3}$$

⑨ A

$$P(2) = 4$$

$$a \times 2^3 - 5 \times 2^2 + 10 \times 2 + 12 = 4$$

$$8a + 12 = 4$$

$$8a = -8$$

$$a = -1$$

$$P(x) = -x^3 - 5x^2 + 10x + 12$$

$$P(-3) = -(-3)^3 - 5(-3)^2 + 10(-3) + 12$$

$$= -36$$

⑩ C

$$y = \cos(\ln x)$$

$$\frac{dy}{dx} = -\sin(\ln x) \times \frac{1}{x}$$

$$= \frac{-\sin(\ln x)}{x}$$

$$\left\{ \begin{array}{l} u = -\sin(\ln x) \\ u' = -\cos(\ln x) \end{array} \right. \quad \left\{ \begin{array}{l} v = x \\ v' = 1 \end{array} \right.$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{-\sin(\ln x)}{x} \right) = \frac{-\cos(\ln x)}{x} - \frac{-\sin(\ln x)}{x^2}$$

$$= \frac{\sin(\ln x) - \cos(\ln x)}{x^2}$$

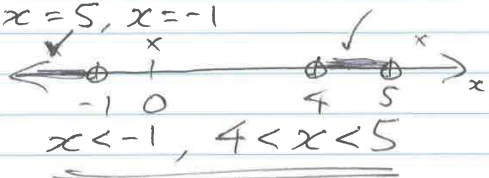
$$\begin{aligned}
 \textcircled{11} \text{ a) } \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\
 &= \left(\frac{-(-5)}{6}\right)^2 - 2\left(\frac{-13}{6}\right) \\
 &= \frac{181}{36} \text{ OR } 5\frac{1}{36}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } P \times 1, A \times 4, R \times 2, M \times 1, T \times 2 \\
 \text{No. arrangements} &= 10! \\
 &= \frac{4! \times 2! \times 2!}{1} \\
 &= \underline{\underline{37800}}
 \end{aligned}$$

c) Critical Points

Equality	Zero Denom
$\frac{5}{x-4} = x$	$x-4=0$
	$x=4$

$$\begin{aligned}
 5 &= x^2 - 4x \\
 x^2 - 4x - 5 &= 0 \\
 (x-5)(x+1) &= 0 \\
 x &= 5, x = -1
 \end{aligned}$$



$$\begin{aligned}
 \text{d) } u &= e^x & x &= \ln \frac{1}{\sqrt{2}} \rightarrow u = e^{\ln(\frac{1}{\sqrt{2}})} = \frac{1}{\sqrt{2}} \\
 du &= e^x dx & x &= 0 \rightarrow u = e^0 = 1
 \end{aligned}$$

$$\begin{aligned}
 \int_{\frac{1}{\sqrt{2}}}^1 \frac{5}{\sqrt{1-u^2}} du &= \left[5 \sin^{-1} u \right]_{\frac{1}{\sqrt{2}}}^1 \\
 &= 5 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) - 5 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \\
 &= 5 \times \frac{\pi}{4} - 5 \times \frac{\pi}{2} \\
 &= \underline{\underline{-\frac{5\pi}{4}}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{11} \text{ e) } \frac{d}{dx} (x^2 \cos^{-1} x) & \quad \begin{cases} u = x^2 & v = \cos^{-1} x \\ u' = 2x & v' = \frac{-1}{\sqrt{1-x^2}} \end{cases} \\
 &= \underline{\underline{2x \cos^{-1} x - \frac{x^2}{\sqrt{1-x^2}}}}
 \end{aligned}$$

$$\text{f) Power ratio required} = 1:3$$

$$= 3:9$$

$$\text{Term independent of } x = {}^{12}C_9 (2x^3)^3 \left(-\frac{3}{x}\right)^9$$

$$= {}^{12}C_9 (2^3)^3 (-3)^9$$

$$= -{}^{12}C_9 (2^3)^3 (3^9) \text{ OR } -{}^{12}C_3 (2^3)^3 (3^9)$$

⑫ a) i) $T = 160 - Ae^{-kt}$

When $t=0$, $T=-20$

$-20 = 160 - Ae^0$

$A = 180$

$\therefore T = 160 - 180e^{-kt}$

When $t=50$, $T=40$

$40 = 160 - 180e^{-50k}$

$180e^{-50k} = 120$

$e^{-50k} = \frac{2}{3}$

$e^{50k} = \frac{3}{2}$

$50k = \ln\left(\frac{3}{2}\right)$

$k = \frac{1}{50} \ln\left(\frac{3}{2}\right)$

ii) $T = 160 - 180e^{-\frac{1}{50} \ln\left(\frac{3}{2}\right) t}$

When $T=60$

$60 = 160 - 180e^{-\frac{1}{50} \ln\left(\frac{3}{2}\right) t}$

$180e^{-\frac{1}{50} \ln\left(\frac{3}{2}\right) t} = 100$

$e^{-\frac{1}{50} \ln\left(\frac{3}{2}\right) t} = \frac{5}{9}$

$t = \frac{\ln\left(\frac{5}{9}\right)}{-\frac{1}{50} \ln\left(\frac{3}{2}\right)}$

$= 72.48 \dots \text{mins}$

$= 72 \text{ OR } 73 \text{ (accept either)}$

b) $P(HH) = \frac{1}{4}$ $P(\text{other}) = \frac{3}{4}$

$\left(\frac{1}{4} + \frac{3}{4}\right)^{10}$

$P = {}^{10}C_6 \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right)^6 \text{ OR } {}^{10}C_4 \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^4$

⑫ c) i) $\angle ABD = \frac{\theta}{2}$ (Angle at centre $2 \times \angle$ at circumference standing on same arc)

$\angle DBY = 180^\circ - \frac{\theta}{2}$ (straight $\angle DBY = 180^\circ$)

ii) Similarly $\angle ACY = 180^\circ - \frac{\theta}{2}$

$180^\circ - \frac{\theta}{2} + 180^\circ - \frac{\theta}{2} + \angle BXC + \angle BYC = 360^\circ$ ($\angle$ sum of quadrilateral BXC Y)

$360^\circ - \theta + \angle BXC + \angle BYC = 360^\circ$

$\therefore \angle BXC + \angle BYC = \theta$

d) $V = \pi \int_0^\pi \left(\cos \frac{x}{2}\right)^2 dx$

$= \pi \int_0^\pi \cos^2 \frac{x}{2} dx$

$= \pi \int_0^\pi \frac{1}{2} + \frac{1}{2} \cos x dx$

$= \pi \left[\frac{x}{2} + \frac{1}{2} \sin x \right]_0^\pi$

$= \pi \left(\left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right) - \left(\frac{0}{2} + \frac{1}{2} \sin 0 \right) \right)$

$= \frac{\pi^2}{2} \text{ units}^3$

(12) e) Prove for $n=2$

$$\text{LHS} = 2^2 = 4 \quad \text{RHS} = 2 \times 2 = 4$$

$$\therefore \text{LHS} \geq \text{RHS}$$

$\therefore$ true for $n=2$

Assume true for $n=k$

i.e. assume $k^2 \geq 2k$

Prove true for $n=k+1$

RTP $(k+1)^2 \geq 2(k+1)$

$$\begin{aligned} \text{LHS} &= (k+1)^2 & \text{RHS} &= 2(k+1) \\ &= k^2 + 2k + 1 & &= 2k + 2 \end{aligned}$$

Now $k^2 + 2k + 1 \geq 2k + 2k + 1$ by assumption

but $2k + 2k + 1 \geq 2k + 2$ as $k \geq 2$

$$\therefore k^2 + 2k + 1 \geq 2k + 2$$

$$\text{LHS} \geq \text{RHS}$$

$\therefore$ True for $n=k+1$

$\therefore$ True by mathematical induction for all integers $n \geq 2$

(13) a) i) $f(x) = \frac{2x^2+9}{x^2+4}$ OR $2 + \frac{1}{x^2+4} = \frac{2(x^2+4)}{x^2+4} + \frac{1}{x^2+4}$

$$\begin{aligned} &= \frac{2(x^2+4) + 1}{x^2+4} & &= \frac{2x^2+8+1}{x^2+4} \\ &= \frac{2(x^2+4)}{x^2+4} + \frac{1}{x^2+4} & &= \frac{2x^2+9}{x^2+4} \\ &= 2 + \frac{1}{x^2+4} & &= f(x) \end{aligned}$$

ii) $y = 2$

iii) $f'(x) = \frac{-2x}{(x^2+4)^2}$

when $f'(x) = 0$

$$-2x = 0$$

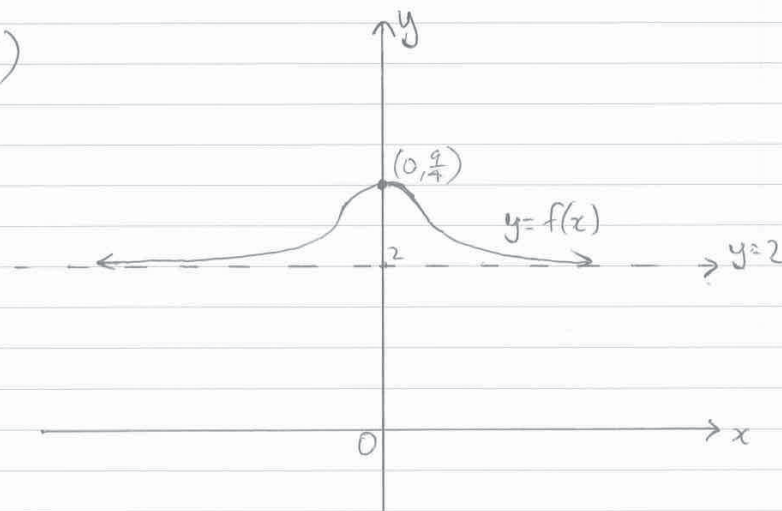
$$x = 0$$

$$f(0) = \frac{2 \times 0^2 + 9}{0^2 + 4}$$

$$= \frac{9}{4}$$

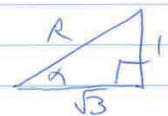
$$\therefore \left(0, \frac{9}{4}\right)$$

iv)



$$\begin{aligned}
 \textcircled{13} \text{ a) v) Area} &= 2 \int_0^2 \left(2 + \frac{1}{x^2 + 4} \right) dx \\
 &= 2 \left[2x + \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \right]_0^2 \\
 &= 2 \left(\left(2(2) + \frac{1}{2} \tan^{-1} \left(\frac{2}{2} \right) \right) - \left(2(0) + \frac{1}{2} \tan^{-1} \left(\frac{0}{2} \right) \right) \right) \\
 &= 2 \left(4 + \frac{1}{2} \times \frac{\pi}{4} \right) \\
 &= 8 + \frac{\pi}{4} \text{ units}^2
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{13} \text{ b) i) } R \sin(2t + \alpha) &= R \sin 2t \cos \alpha + R \cos 2t \sin \alpha \\
 \therefore R \cos \alpha &= \sqrt{3} & R \sin \alpha &= 1 \\
 \cos \alpha &= \frac{\sqrt{3}}{R} & \sin \alpha &= \frac{1}{R} \\
 R^2 &= 1^2 + (\sqrt{3})^2 & \alpha &= \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \\
 R &= \sqrt{4} & &= \frac{\pi}{6} \\
 &= 2 \\
 \therefore x &= 2 \sin \left(2t + \frac{\pi}{6} \right)
 \end{aligned}$$



$$\begin{aligned}
 \text{ii) } \dot{x} &= 4 \cos \left(2t + \frac{\pi}{6} \right) \\
 \ddot{x} &= -8 \sin \left(2t + \frac{\pi}{6} \right) \\
 &= -4 \times 2 \sin \left(2t + \frac{\pi}{6} \right) \\
 &= -4x \\
 &= -2^2 x \quad \therefore \text{In form } \ddot{x} = -n^2 x \text{ with } n=2
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) When } \dot{x} &= 0 \\
 0 &= 4 \cos \left(2t + \frac{\pi}{6} \right) \\
 2t + \frac{\pi}{6} &= \frac{\pi}{2} \\
 2t &= \frac{\pi}{3} \\
 t &= \frac{\pi}{6} \text{ sec} \\
 x &= 2 \sin \left(2 \times \frac{\pi}{6} + \frac{\pi}{6} \right) \\
 &= 2
 \end{aligned}$$

$$\text{iv) Period} = \pi \text{ and } a = 2$$



(It goes through a full cycle in π seconds so the $\frac{\pi}{6}$ is irrelevant for distance)

$$\begin{aligned}
 \text{Distance travelled} &= 4 \times 2 \\
 &= \underline{8 \text{ m}}
 \end{aligned}$$

14 a) Let $f(x) = x^3 - \cos x$
 $f'(x) = 3x^2 + \sin x$
 $x_2 = 0.9 - \frac{0.9^3 - \cos 0.9}{3(0.9)^2 + \sin 0.9}$
 $= 0.8665...$
 $\div 0.87$

b) $4 \sin \frac{x}{2} \cos \frac{x}{2} = \sqrt{3}$
 $2 \sin \frac{x}{2} \cos \frac{x}{2} = \frac{\sqrt{3}}{2}$
 $\sin x = \frac{\sqrt{3}}{2}$
 $x = n\pi + (-1)^n \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$
 $= n\pi + (-1)^n \frac{\pi}{3}$ where n is an integer

c) $y^2 = 13^2 - x^2$
 $y = \sqrt{169 - x^2}$
 $\frac{dy}{dx} = \frac{1}{2}(169 - x^2)^{-\frac{1}{2}} \cdot -2x$ $\frac{dx}{dt} = -0.1$
 $= \frac{-x}{\sqrt{169 - x^2}}$
 $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$
 $= \frac{-5}{\sqrt{169 - 5^2}} \times -0.1$
 $= \frac{1}{24} \text{ ms}^{-1}$

14 d) i) $v = 25$, $g = 9.8$
 $x = 25t \cos \theta$ $y = -\frac{1}{2} \times 9.8 \times t^2 + 25t \sin \theta$
 $\rightarrow t = \frac{x}{25 \cos \theta}$ ① $= -4.9t^2 + 25t \sin \theta$ ②
 Sub ① into ②
 $y = -4.9 \left(\frac{x}{25 \cos \theta} \right)^2 + 25 \left(\frac{x}{25 \cos \theta} \right) \sin \theta$
 $= \frac{-4.9x^2}{625 \cos^2 \theta} + x \tan \theta$
 $= \frac{-49x^2}{6250} \sec^2 \theta + x \tan \theta$
 $= \frac{-49x^2}{6250} (1 + \tan^2 \theta) + x \tan \theta$

ii) Let $x = 60$ & $y = 1.2$
 $1.2 = \frac{-49(60)^2}{6250} (1 + \tan^2 \theta) + 60 \tan \theta$
 $1.2 = \frac{-3528}{125} - \frac{3528}{125} \tan^2 \theta + 60 \tan \theta$
 $\frac{3528}{125} \tan^2 \theta - 60 \tan \theta + \frac{3678}{125} = 0$
 $3528 \tan^2 \theta - 7500 \tan \theta + 3678 = 0$
 $1764 \tan^2 \theta - 3750 \tan \theta + 1839 = 0$
 $\tan \theta = \frac{3750 \pm \sqrt{(-3750)^2 - 4(1764)(1839)}}{2 \times 1764}$
 $= 1.3583... , 0.767...$
 $\theta = 53.64...^\circ , 37.505...^\circ$
 $\div 53^\circ 38' , 37^\circ 30'$
 $\therefore 37^\circ 30' < \theta < 53^\circ 38'$
 (Accept $38^\circ < \theta < 54^\circ$)

$$\begin{aligned}
 (14) \text{ e) i) } & {}^nC_0 x^{-1} + {}^nC_1 + {}^nC_2 x + {}^nC_3 x^2 + {}^nC_4 x^3 + \dots + {}^nC_n x^{n-1} \\
 &= \frac{{}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + {}^nC_4 x^4 + \dots + {}^nC_n x^n}{x} \\
 &= \frac{(1+x)^n}{x}
 \end{aligned}$$

$$\text{ii) } \frac{(1+x)^n}{x} = {}^nC_0 x^{-1} + {}^nC_1 + {}^nC_2 x + {}^nC_3 x^2 + {}^nC_4 x^3 + \dots + {}^nC_n x^{n-1}$$

Differentiating both sides

$$\frac{x \cdot n(1+x)^{n-1} - (1+x)^n}{x^2} = -{}^nC_0 x^{-2} + {}^nC_2 + 2{}^nC_3 x + 3{}^nC_4 x^2 + \dots + (n-1){}^nC_n x^{n-2}$$

When $x=1$

$$n(2)^{n-1} - 2^n = -{}^nC_0 + {}^nC_2 + 2{}^nC_3 + 3{}^nC_4 + \dots + (n-1){}^nC_n$$

$$n(2)^{n-1} - 2 \times 2^{n-1} = -1 + {}^nC_2 + 2{}^nC_3 + 3{}^nC_4 + \dots + (n-1){}^nC_n$$

$$2^{n-1}(n-2) = {}^nC_2 + 2{}^nC_3 + 3{}^nC_4 + \dots + (n-2){}^nC_{n-1} + (n-2){}^nC_n$$

$$(n-2)(2^{n-1} - 1) = {}^nC_2 + 2{}^nC_3 + 3{}^nC_4 + \dots + (n-2){}^nC_{n-1}$$